

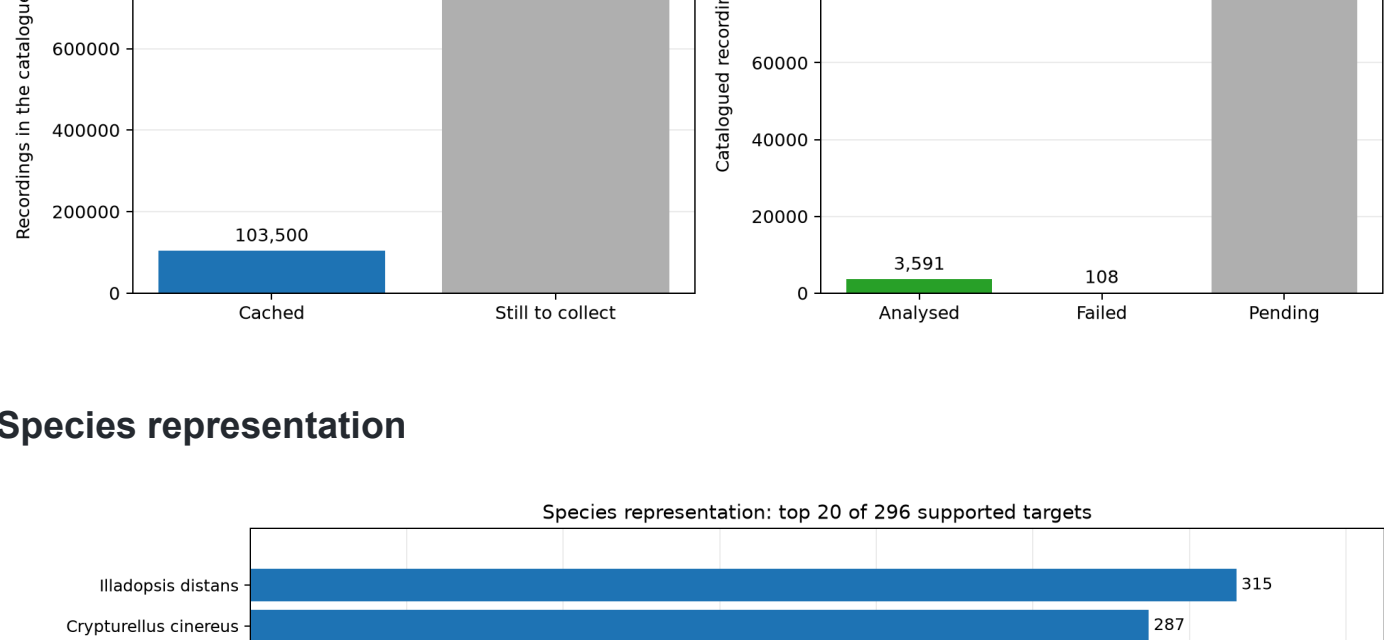
Xeno-canto robustness results

This notebook displays cumulative measurements from Xeno-canto recordings.

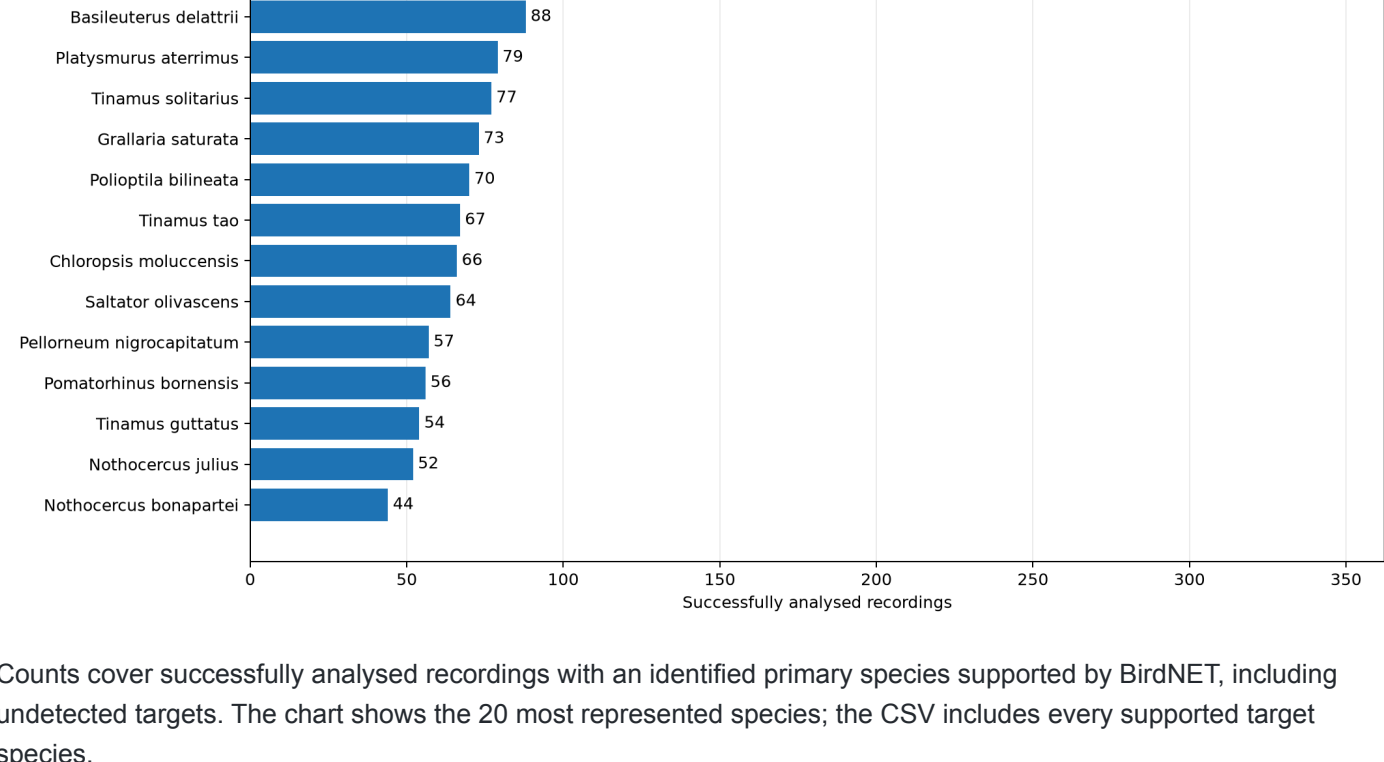
Data and coverage

3,591 recordings analysed · 108 failures · 7 completed batches · cursor 3,699

Corpus progress

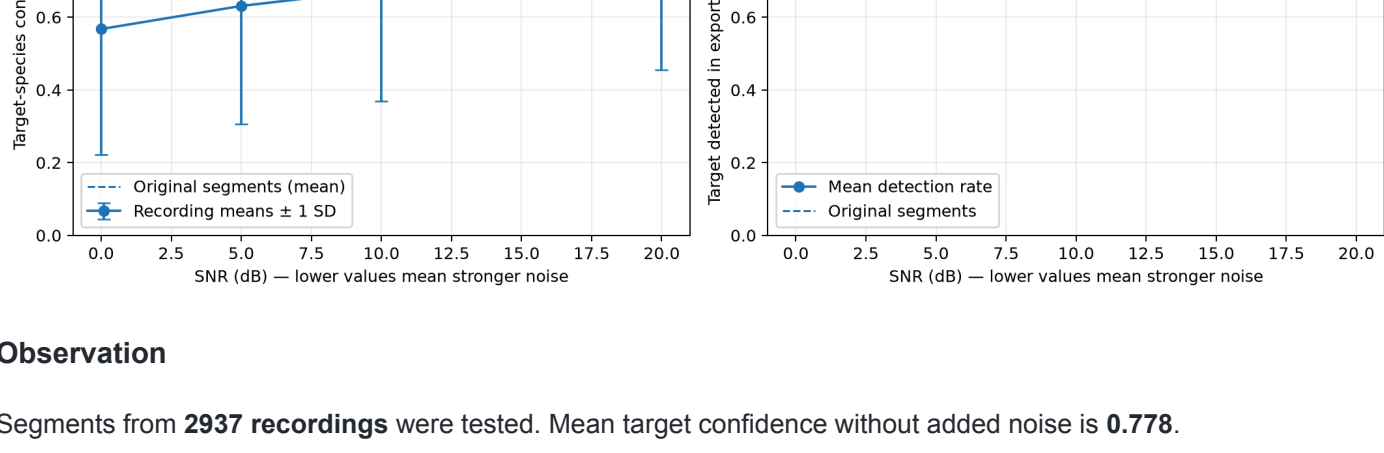


Species representation



Counts cover successfully analysed recordings with an identified primary species supported by BirdNET, including uncovered targets. The chart shows the 20 most represented species; the CSV includes every supported target species.

Noise robustness

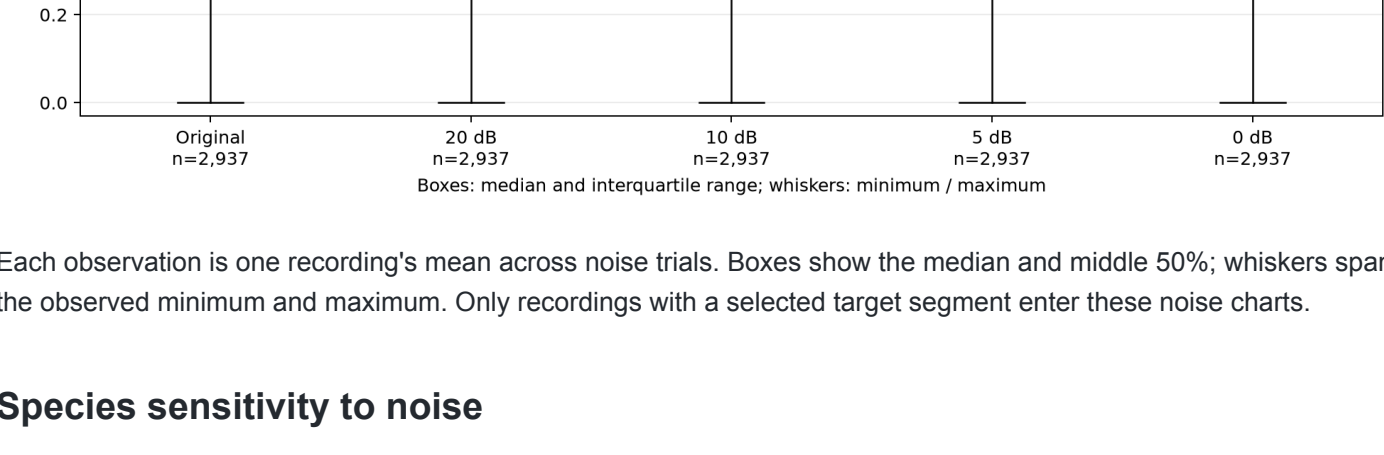


Observation

Segments from 2937 recordings were tested. Mean target confidence without added noise is 0.778.

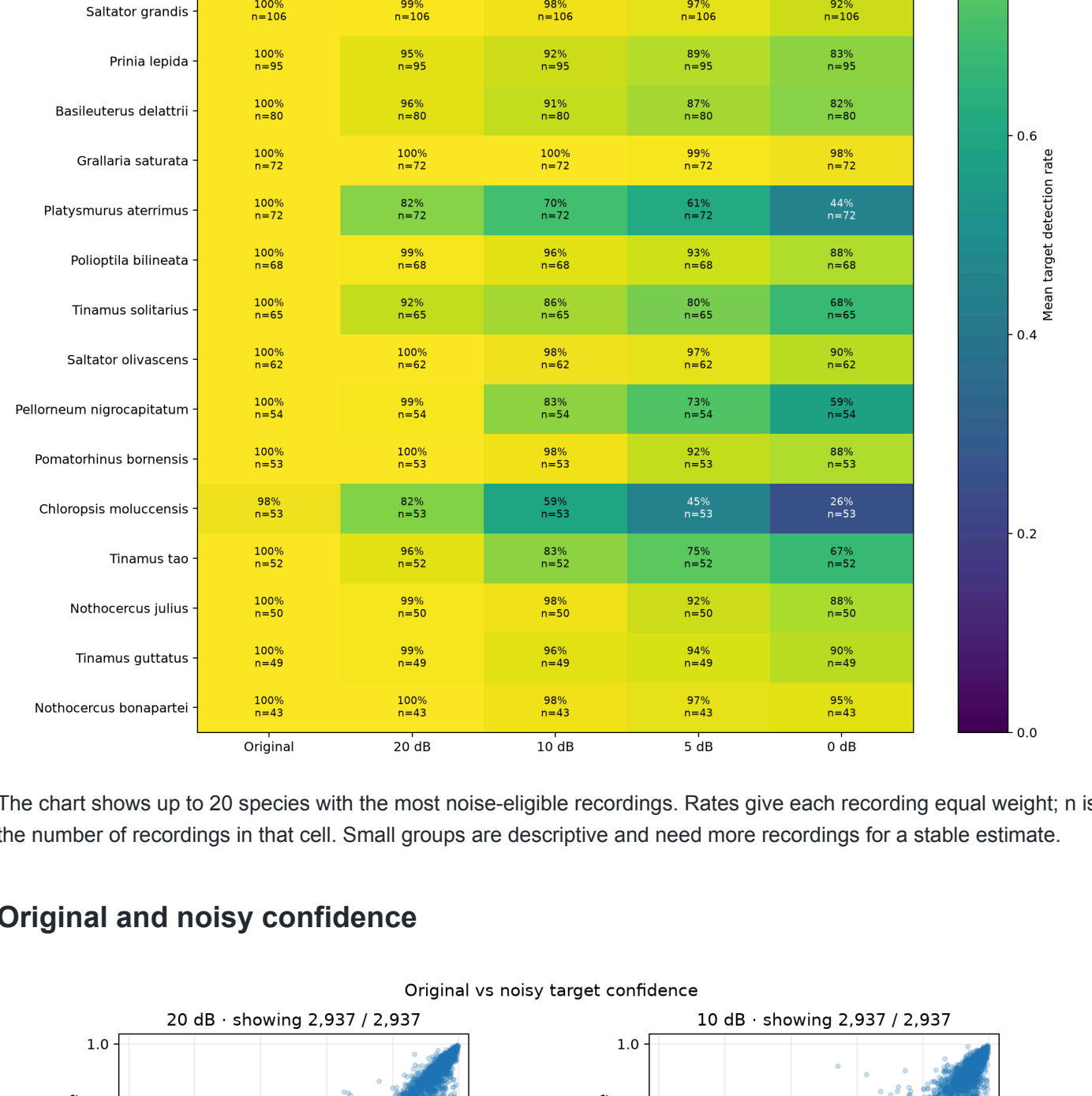
- **20 dB**: mean confidence 0.731, standard deviation across recordings 0.277, target detected in 96.6% of trials.
 - **10 dB**: mean confidence 0.676, standard deviation across recordings 0.307, target detected in 92.2% of trials.
 - **5 dB**: mean confidence 0.631, standard deviation across recordings 0.325, target detected in 88.5% of trials.
 - **0 dB**: mean confidence 0.568, standard deviation across recordings 0.346, target detected in 82.5% of trials.
- Error bars show one standard deviation of recording means. A value of 0 means the target is absent from predictions exported at the minimum confidence threshold of 0.10. 654 recordings without an initially detected target were excluded from this test.

Confidence distributions by noise level



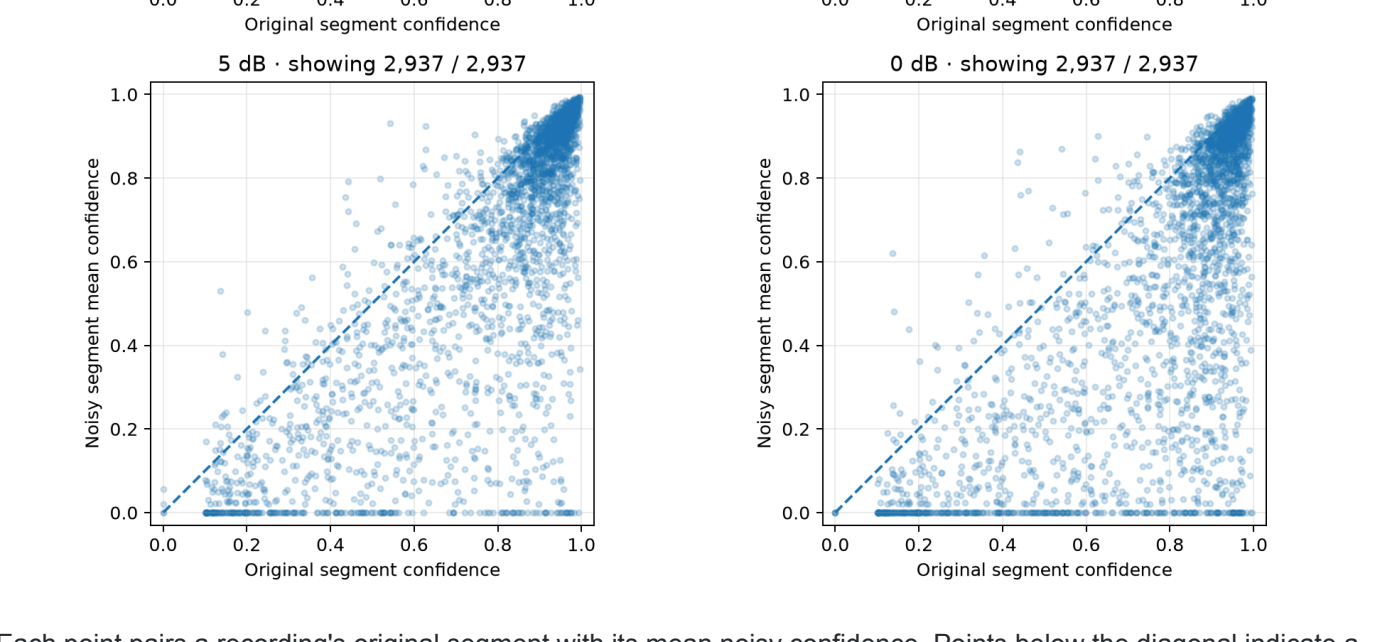
Each observation is one recording's mean across noise trials. Boxes show the median and middle 50%; whiskers span the observed minimum and maximum. Only recordings with a selected target segment enter these noise charts.

Species sensitivity to noise



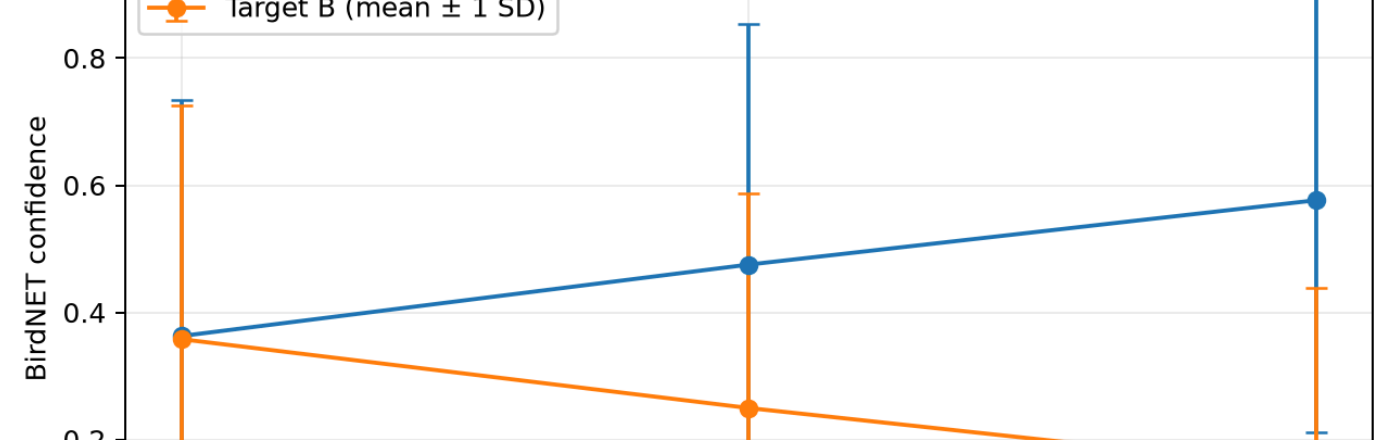
The chart shows up to 20 species with the most noise-eligible recordings. Rates give each recording equal weight; n is the number of recordings in that cell. Small groups are descriptive and need more recordings for a stable estimate.

Original and noisy confidence



Each point pairs a recording's original segment with its mean noisy confidence. Points below the diagonal indicate a decrease. Up to 5,000 recordings per condition are selected deterministically across processing order. Confidence is not a measure of accuracy.

Overlapping vocalizations

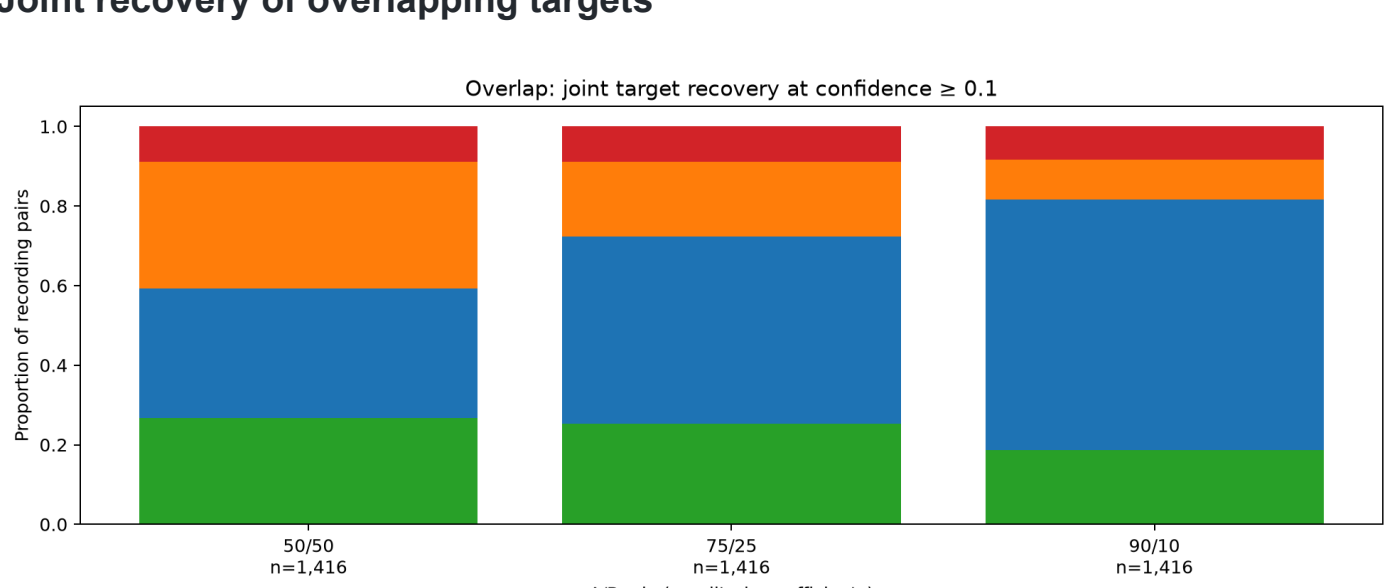


Observation

Means cover 1416 pairs of distinct recordings. Between mixes 50/50 and 90/10, mean target A confidence changes from 0.363 to 0.576, and target B confidence from 0.358 to 0.153.

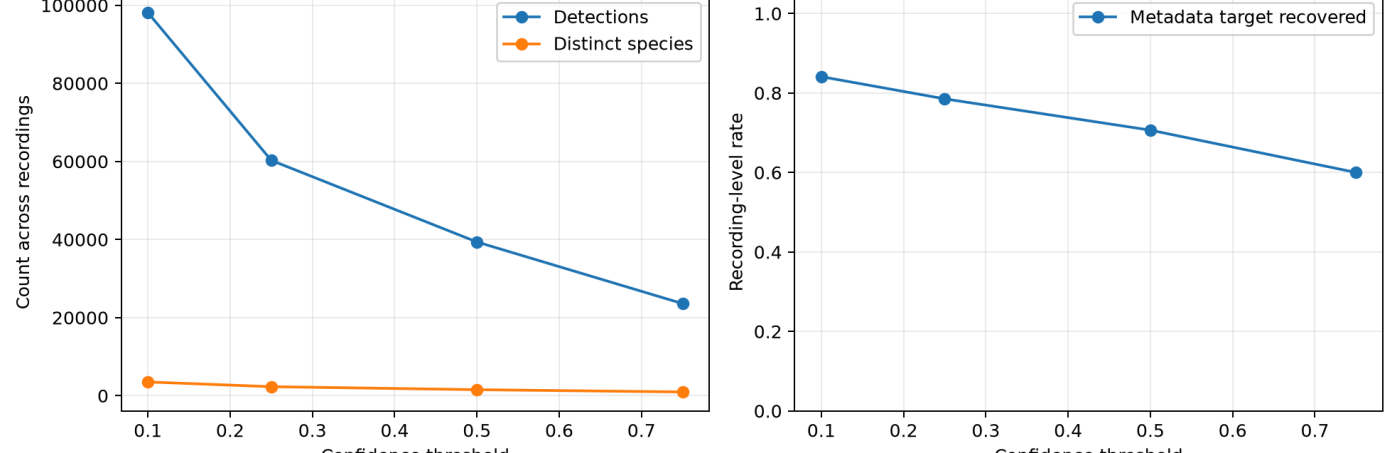
A and B identify the roles within each pair. Error bars show one standard deviation across pairs. Coefficients apply to amplitude, without prior level equalization.

Joint recovery of overlapping targets



The four outcomes use the minimum confidence threshold for the experiment. Each mixture counts a pair once; n is its pair count. A and B describe roles within each pair.

Confidence threshold



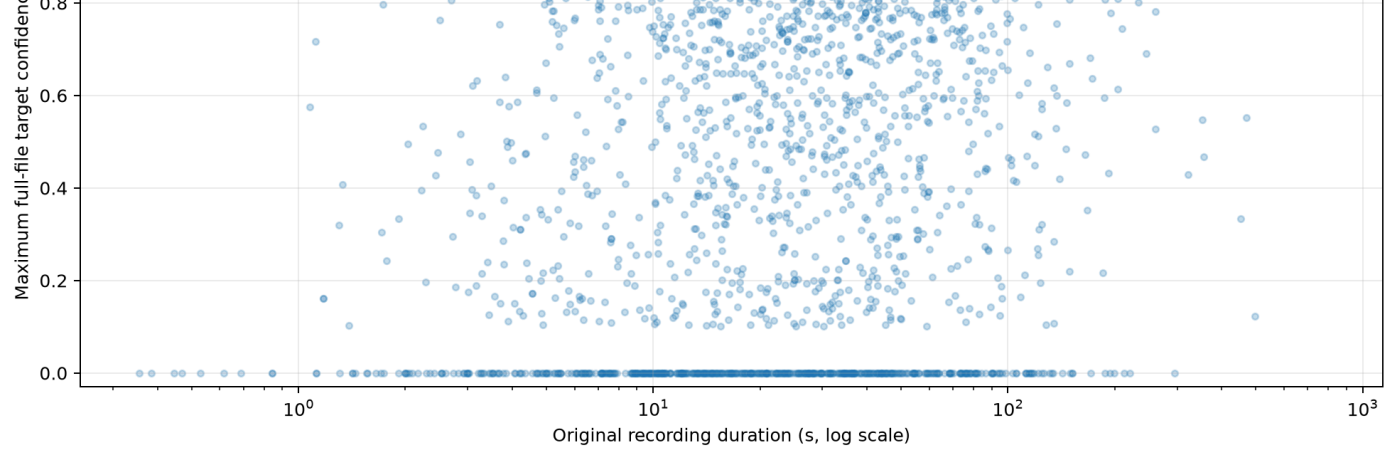
Observation

Between thresholds 0.10 and 0.75, retained detections change from 98117 to 23583, and predicted species from 3519 to 960.

Xeno-canto's primary species is recovered in 84.1% then 60.0% of recordings whose target is identified and supported by BirdNET. This rate measures recovery of the known target; recall across all species requires complete annotations.

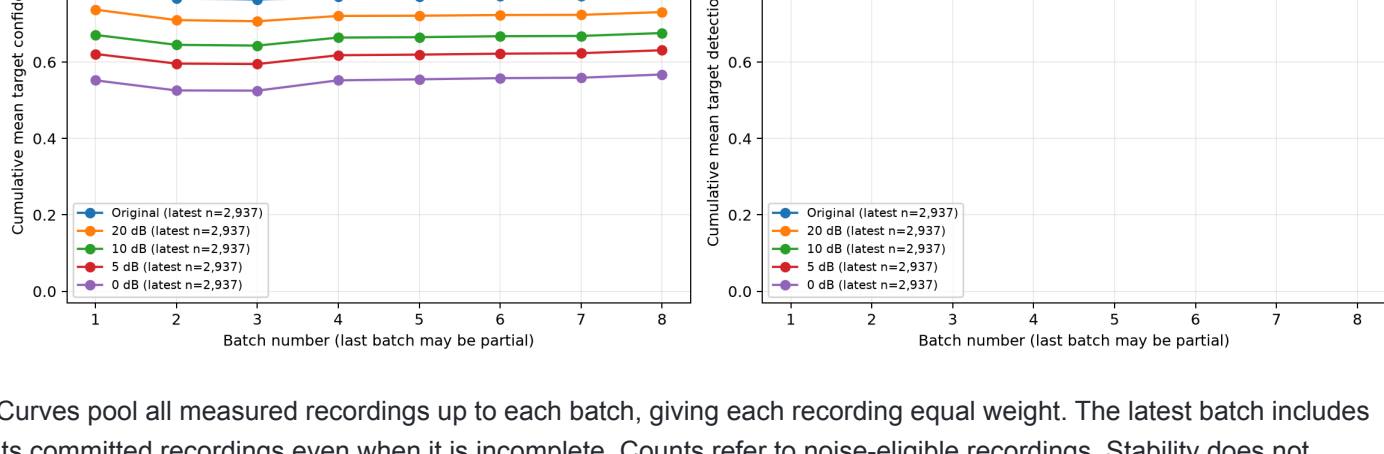
Precision, recall and F1 are not calculated without complete annotations.

Recording duration and target confidence



Each point relates the original recording's duration to the maximum exported confidence for its supported primary species. Zero indicates that the target was absent from exported predictions. Up to 5,000 recordings are selected across processing order. Longer recordings offer more windows for a maximum; this plot describes an association rather than a causal effect.

Cumulative estimates across batches



Curves pool all measured recordings up to each batch, giving each recording equal weight. The latest batch includes its committed recordings even when it is incomplete. Counts refer to noise-eligible recordings. Stability does not establish that the processed subset represents the complete catalogue.

Spectrogram

Example of a short analysed segment.

